Engine Control ECU - HLD v1

Title and Document Control

SYNTHETIC DATA FOR ACADEMIC PROJECT

Engine Control ECU - High Level Design

Document Version: v1

Engine Components

EngineControlSWC manages engine speed, engine state and engine control command processing. SensorManagerSWC supplies normalized engine sensor information.

Interfaces

EngineSpeedInterface provides EngineSpeed and EngineSpeedStatus. EngineCommandInterface carries EngineCommand. EngineControlSWC requires VehicleSpeedInterface and provides EngineSpeedInterface.

Dependencies

EngineControlSWC depends on SensorManagerSWC for sensor input and VehicleControlSWC for approved engine command requests.

Communication

EngineSpeedInterface is transported on CAN0. EngineCommandInterface is transported on CAN0.

Diagnostics

DiagnosticManagerSWC reads engine health data from EngineControlSWC. DiagnosticGatewaySWC routes diagnostic traffic toward the engine-control environment.